

Prathmesh Shelke

+91 (784) 192 8682 | shelkeprathmesh587@gmail.com | prathmesh3245.github.io/profim/ | github.com/prathmesh3245 | linkedin.com/in/prathmesh-shelke-22389b25a/

AI Engineer & Full-Stack Developer | Building Scalable Apps with Intelligent Backends | Open to Internship Challenges

EDUCATION

Savitribai Phule Pune University | *BE in AI & Data Science* | 2022 – 2026 | GPA: **9.05 / 10.0** Jan 2025 - Present
Tech Finalists: Medecro Healthcare Hackathon & Bachelor's *Real-time sensor data pipeline* in AI & Data Science Department
Courses: Programming with Python | Data Structures & Algorithms | C & C++ (OOPs) | Statistics | Data Science | Ai and Fundamentals | SQL | MongoDB | MERN | Generative AI on Diffusion Models | Flask

EXPERIENCE

IGT Solutions, Gen AI Intern | Remote (Pune, India)

Jan 2025 - Present

- Reduced query response time by 40% using ChromaDB indexing. Accelerated development by 40%, delivered a fully functional AI-controlled robot from scratch.
- Built custom AI chatbot using AI21's Jamba LLM and ChromaDB, enabling semantic search over 5+ PDF documents.
- Optimized retrieval pipeline, reducing response latency via vector indexing and caching.
- I played a critical role in an agile team of 5 developers to design and implement a robust application in 2 months. This increased the workplace productivity by 25%.

Medecro HealthCare Hackathon, (Team Lead) | Remote (India)

||Aug 2024 - Sept 2024

Developed an AI-powered medical dashboard with real-time disease prediction and diagnostic tools.

- Led a 3-person team in building a full-stack medical dashboard integrating predictive analytics for early disease detection.
- Spearheaded UI/UX development for real-time patient data visualization, enabling doctors to track trends and anomalies efficiently.
- **Advanced to finals** among 100+ teams, recognized for innovation in healthcare AI.

Smart India Hackathon, Frontend Developer (Team Lead) | India

Aug 2024 – Sept 2024

- Led a 6-person cross-functional team to build a diagnostic tool with real-time disease prediction using ML and interactive dashboards.
- Automated evaluation pipelines with **rule-based scripting** and rigorous unit testing, achieving **95% code coverage**.
- **Selected at college level** among 50+ teams for innovation in healthcare tech; advanced to national problem statement pool.

AESA, Technical Team (Team Member, AI & DS Department) | India

July 2023 - March 2024

- Organized a hands-on competitive coding and problem-solving event to promote technical skills among 50+ participants.
- **Designed problem statements** and scoring criteria, ensuring a balance of difficulty for beginners and advanced coders.
- **Managed logistics**, including platform setup (HackerRank/CodeChef), judging, and mentorship for participants.
- **Led marketing efforts**: Designed promotional content (social media graphics, emails, posters) and outreach campaigns, increasing registrations by **40%** over previous events.

SKILLS

Languages	C, C++, Javascript, Python, React, Express, Node.js, SQL, Git, Bash, Flask
Software	Linux, MongoDB Compass, Postman, Claude (MCP Server), Alpaca (Paper Trading API), GitHub, MySQL CLI, AI21 Labs.
Certifications	Data Structures & Algorithms, C & C++ (OOPs), SQL, Generative AI with Flask.
AI/ML	Python, PyTorch, TensorFlow, ChromaDB, Hugging Face.
Data Analysis	Python, Statistics, MS-SQL, Microsoft PowerBI, DAX.

PROJECTS

AI21: Document Embedding and AI integration

Jan 2025 - Mar 2025

- Reduced query response time by 40% using **vector** indexing or Handled X GB of document embeddings.
- using AI21’s **Jamba LLM** and **ChromaDB**, enabling semantic search over 500+ PDF documents.
- Optimized retrieval pipeline, reducing response latency via vector indexing and caching.

Expense Tracking and Visualization-

May 2025 -Present

- Developed a high-performance, **low-level** expense tracking system with required authentication using **JSON WEB** Tokens.
- Visualized the expense using **Chart.js** of Javascript and frontend using **React.js**. Backend done using **Express routes**.

Real-time sensor data pipeline

Jan 2023 - April 2023

- Integrated **Arduino-UNO** and performance enhancements led to a 99.9% success rate boost in water level detection.
- Engaged sensors like soil-moisture, **DHT-11** with Relay-shaft along with the processor to take numeric parameters of soil such as temperature, humidity, moisture-level with precision.